

Minseo Kwon

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Research Interests

Robotics, Task and Motion Planning, Learning for Planning

Education

Ewha Womans University, Seoul, Korea

M.S., Artificial Intelligence and Software (Computer Science and Engineering) Sept 2024 – Aug 2026

- Advisor: [Dr. Young. J. Kim](#)
- Current GPA: 4.3/4.3

B.S., Computer Science and Engineering & Mathematics Mar 2020 – Aug 2024

- Graduated with Honors: Magna Cum Laude (GPA: 3.82/4.3)

Publications

Conference Papers

[C02] [M. Kwon](#) and Y. J. Kim, **Kinodynamic Task and Motion Planning using VLM-guided and Interleaved Sampling**, *IEEE International Conference on Robotics and Automation (ICRA)*, 2026, Under Review.

[C02] [M. Kwon](#), Y. Kim, and Y. J. Kim, **Fast and Accurate Task Planning using Neuro-Symbolic Language Models and Multi-level Goal Decomposition**, *IEEE International Conference on Robotics and Automation (ICRA)*, 2025 [🔗](#) [📄](#)

[C01] S. Kwon*, [M. Kwon*](#), H. Kim* and J. Sim, **ToMato: Accelerating ViT via Token Merging**, *The Institute of Electronics and Information Engineers Conference*, 2023. (* Equal Contribution) [🔗](#) [📄](#)

Journal Papers

[J02] Many Authors, **A Dataset and Benchmark for Robotic Cloth Unfolding Grasp Selection: The ICRA 2024 Cloth Competition**, *The International Journal of Robotics Research*, 2025, Under Review. [🔗](#)

[J01] [M. Kwon](#), and Y. Kim, **Neuro-Symbolic Task Replanning using Large Language Models**, *The Journal of Korea Robotics Society*, 2025. [🔗](#)

Research Experience

Ewha Womans University, Seoul, Korea

Computer Graphics Lab, Research Assistant (Advisor: [Dr. Young. J. Kim](#)) Sept 2024 – Present

- **Task and Motion Planning:** Developed a kinodynamic task and motion planning framework that interleaves symbolic planning with motion validation, leveraging a VLM for guided exploration and strategic backtracking, and using physics simulation to enforce physical constraints.
- **WBCD Challenge:** Selected as a finalist (4th place) in the Table Service Operations track in [ICRA 2025 WBCD Challenge](#) among 74 teams.

Computer Graphics Lab, Undergraduate Researcher (Advisor: [Dr. Young. J. Kim](#)) Dec 2022 – Aug 2024

- **Task Planning:** Developed a neuro-symbolic task planner for large-scale robotic tasks, reducing planning time and improving accuracy in various PDDL domains by generating subgoals with multimodal LLM. [\[C02\]](#) [\[J01\]](#)

- **Cloth Manipulation:** Developed a robotic cloth unfolding pipeline leveraging a point cloud-based edge detection method for grasp pose localization, winning 3rd place at the [ICRA 2024 Cloth Competition](#). [\[H03\]](#) 

Capstone Design Project (Advisor: [Dr. Jaehyeong Sim](#))

Jan 2023 – Dec 2023

- **Model Compression:** Accomplished a 22.19% reduction in inference latency for Vision Transformer-based models by recursively merging tokens at the early transformer block while preserving over 80% accuracy. [\[H01\]](#), [\[H02\]](#) [\[C01\]](#)

Teaching Experience

[Altu-Bitu](#), Algorithm Tutoring Program, *Ewha*

Fall 2022

- Conducted lectures on data structures and computer algorithms for 40+ undergraduate students and provided feedback on assignment codes.

Honors & Awards

[H03] [3rd Place](#) | Robotic Grasping of Manipulation Competition (Cloth Manipulation Track), ICRA, 2024

[H02] [Silver Prize](#) | Ewha Engineering Capstone Design Contest, 2023

[H01] [3rd Place](#) | Undergrad Research Paper Contest, Autumn Annual Conference of IEIE, 2023

Scholarships & Academic Awards

[S03] [Graduate Admissions Scholarship \(full tuition for 1 year\)](#) | Ewha, 2024 - 2025

[S02] [Dean's List \(7 semesters\)](#) | Ewha, Fall 2020 - Spring 2024

[S01] [Undergraduate Admissions Scholarship \(full tuition for 4 years\)](#) | Ewha, 2020 - 2023

- Top-ranked entrant among all science and engineering applicants; received full tuition scholarship.

Skills

Programming Languages/Libraries: C++, C, Java, Python, PyTorch, Matlab

Robot SW: ROS, OMPL, MoveIt!, CoppeliaSim, Mujoco, Gazebo, Rviz, Genesis

Robot HW: UR5e, Robotiq 3F adaptive gripper

Languages: Korean (Native), English (Advanced)

Other Activities

EDOC (Ewha Do Coding), Club President

Jan 2022 - Dec 2022

- Led the on-campus algorithm club as president, organizing study sessions, inter-club collaborations, and programming competitions.